

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	B8
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Explainable semantic plagiarism detection

Abstract

Plagiarism detection plays a vital role in educational institutions and research organizations by ensuring the originality and integrity of assignments, reports, academic documents, and research papers. With the increasing availability of digital content and online resources, plagiarism has become more sophisticated, making it difficult to detect using traditional methods. Most existing plagiarism detection systems primarily rely on exact word or phrase matching, which limits their ability to identify paraphrased plagiarism, where the wording is modified while the original meaning is preserved. Furthermore, many existing systems generate only a similarity percentage without providing clear explanations for why the content has been identified as plagiarized, making the results difficult for users to interpret and verify.

This project proposes an Explainable Semantic Plagiarism Detection framework that detects plagiarism based on the semantic meaning of text rather than exact word matching. The proposed system first preprocesses the input documents by extracting and cleaning the text. The documents are then divided into sentences, and each sentence is converted into a semantic representation using the pretrained Sentence-BERT model. Cosine similarity is used to compare the semantic similarity between sentences from different documents. When the similarity score exceeds a predefined threshold, the system identifies the content as potential semantic plagiarism. To improve transparency and user confidence, the framework highlights the semantically matched sentences, displays their similarity scores, identifies the corresponding matching text, and provides a simple explanation indicating that the sentences convey similar meanings despite using different words or sentence structures.

The proposed framework will be evaluated using the PAN Plagiarism Corpus, a publicly available benchmark dataset for plagiarism detection research. The performance of the proposed system will be measured using standard evaluation metrics such as Precision, Recall, F1-Score, and Accuracy, and the results will be compared with conventional text similarity approaches. The expected outcome of this research is an explainable plagiarism detection system capable of identifying both direct and paraphrased plagiarism while presenting understandable evidence to support its predictions. The proposed framework aims to improve the transparency, reliability, and usability of plagiarism detection systems, thereby assisting students, educators, researchers, and academic institutions in making more informed decisions during the plagiarism verification process.

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References (Category specific)

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Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1.	23BQ1A4270	JYESTA TANUJA	
2.	23BQ1A42B4	MUNAGALA JAGADEESWAR	
3.	23BQ1A42C6	NARNE BHAVISHYA	

Project Guide

Name: MR.B.PARDHA SARADHI

Signature:

Project Coordinator

HOD-CSE (AI)